

# Can Increasing COE Quotas Stabilise COE Prices?

Forecasting COE prices and quantifying quota elasticity for Categories A & B

## EXECUTIVE SUMMARY

While COE prices are influenced by multiple market factors, our analysis shows that quota changes have a measurable negative effect on prices.

The proposed model enables LTA to forecast future COE prices and simulate the impact of quota adjustments before policy implementation.



COE prices remain volatile



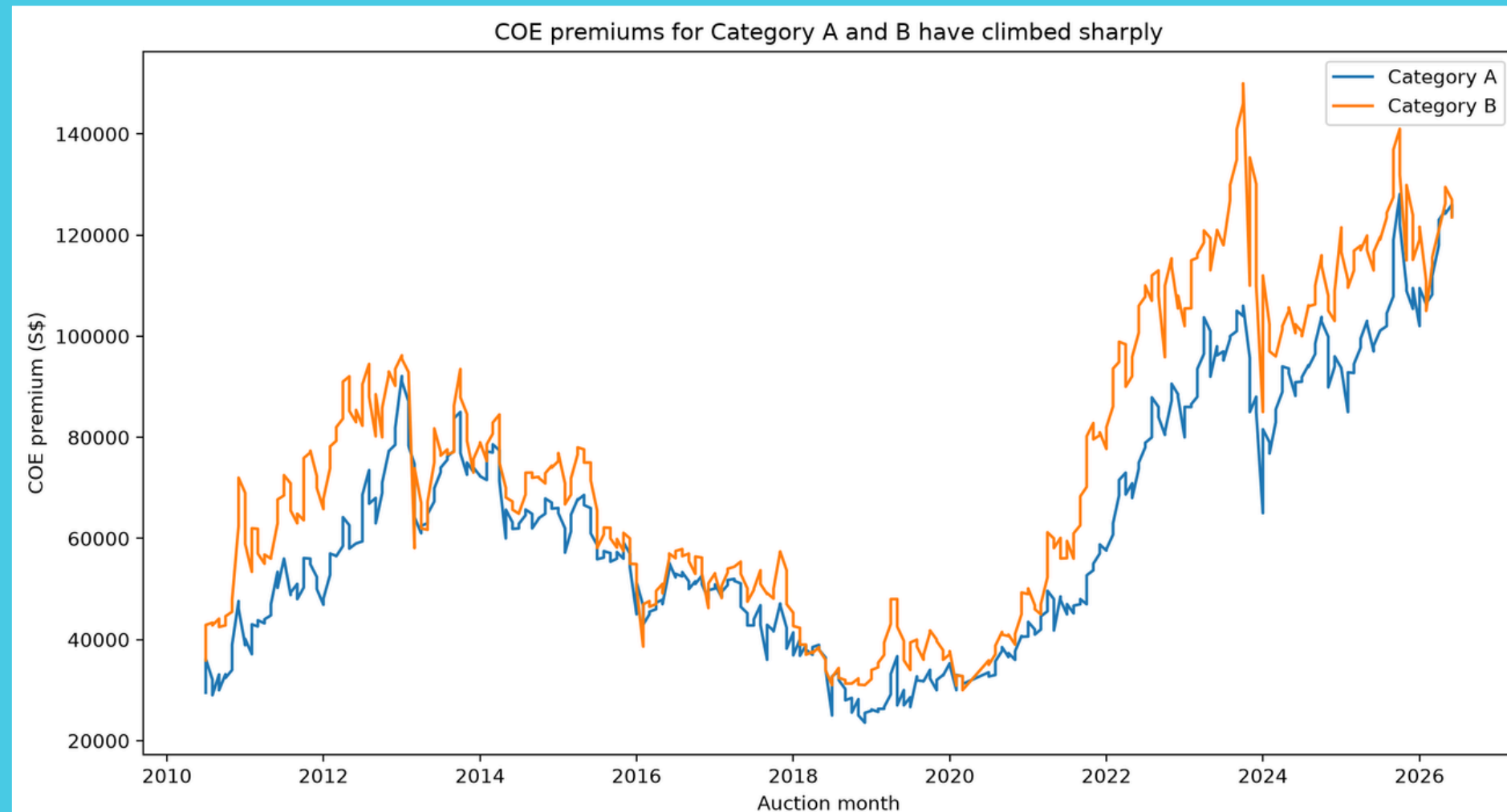
Higher quotas generally reduce prices



Model supports policy simulation



# COE prices have become increasingly volatile

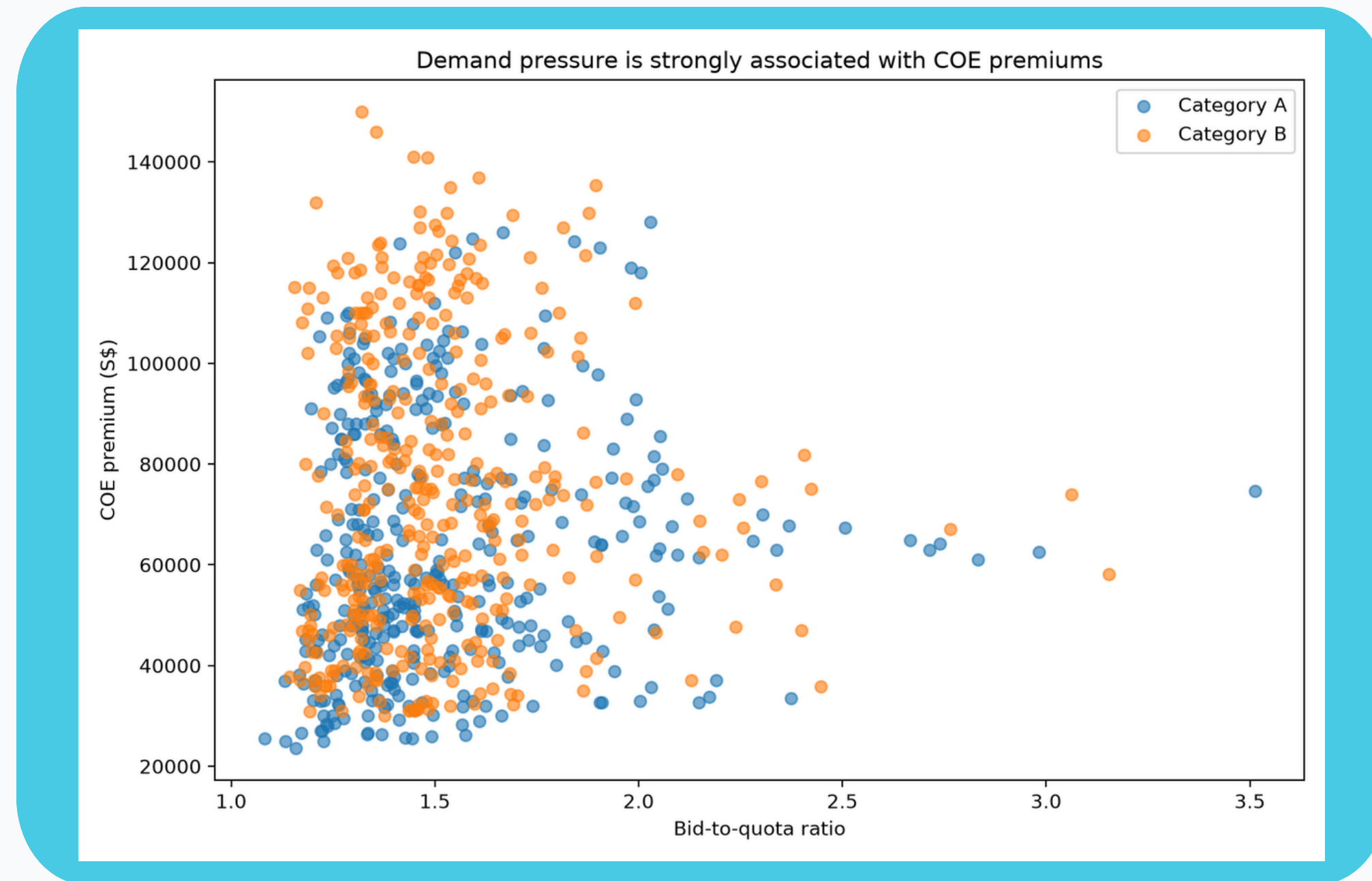


## Key Observations

- Both Category A and B premiums have increased substantially since 2021.
- Category B consistently commands higher premiums than Category A.
- Prices remain highly volatile despite quarterly quota reviews.

The persistence of high and volatile premiums motivates a closer examination of quota policy.

# Higher bidding pressure is associated with higher COE premiums

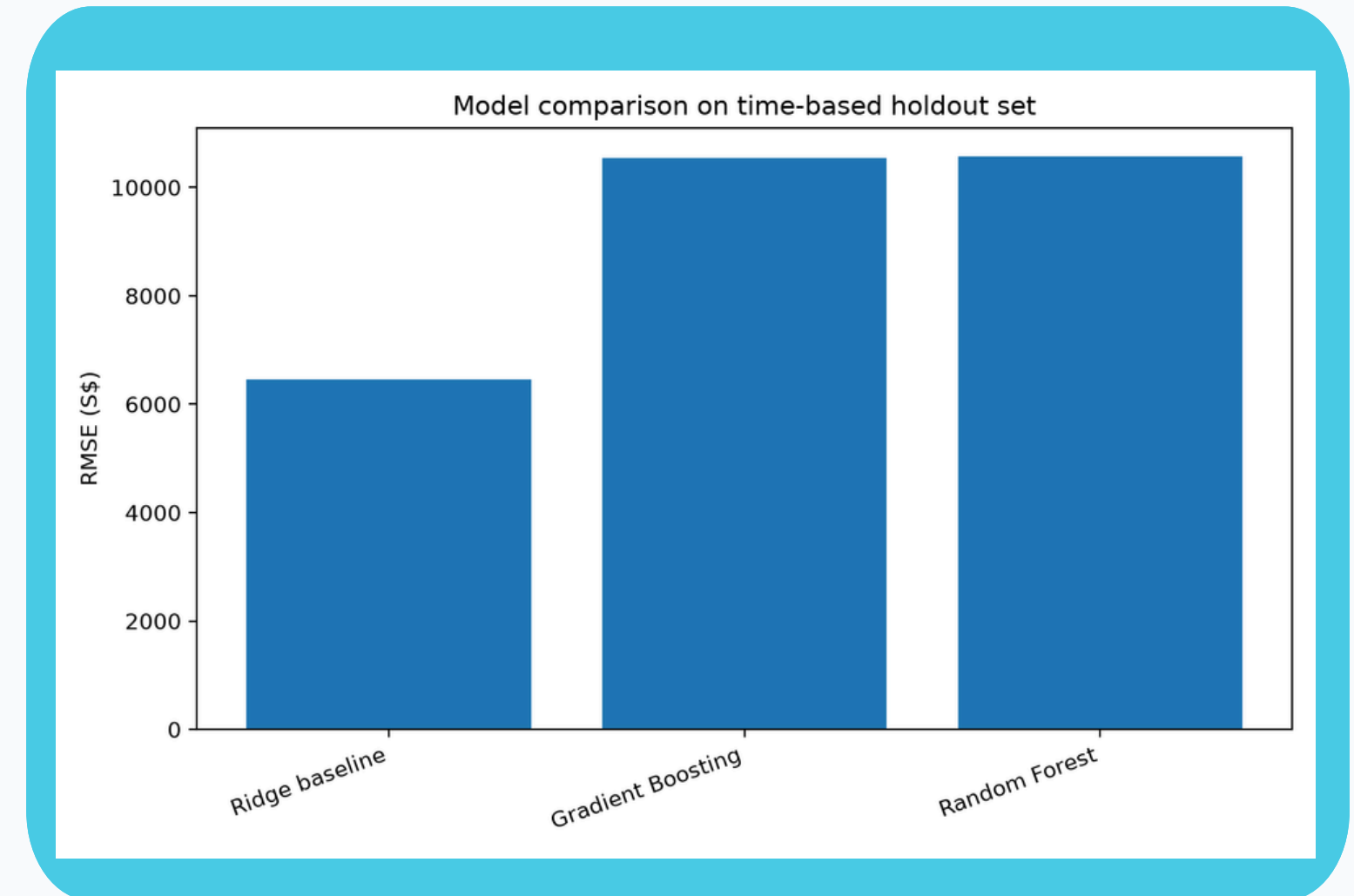
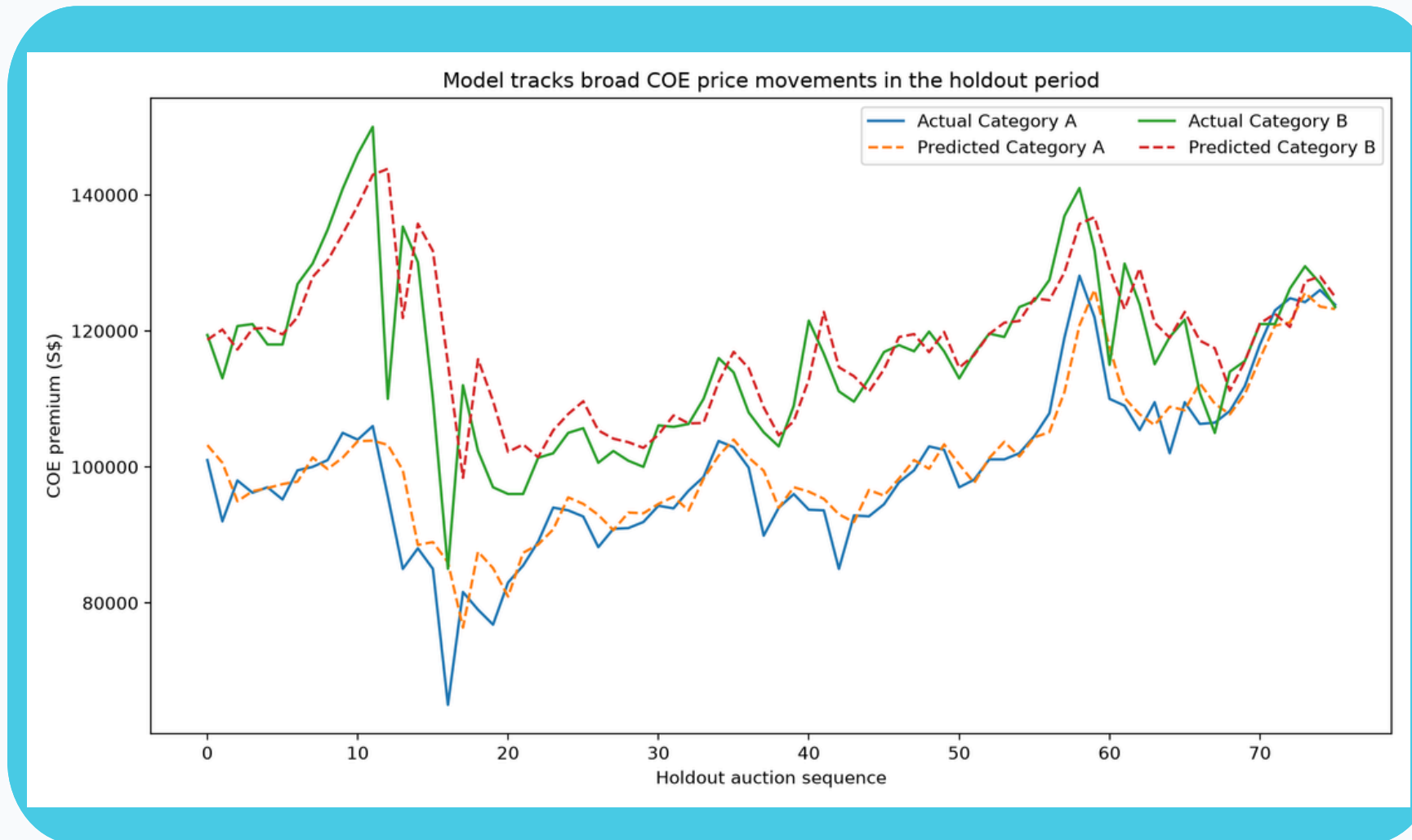


## Key Observations

- Higher bid-to-quota ratios generally correspond to higher premiums.
- Category B exhibits greater price dispersion.
- Demand pressure plays an important role in determining COE prices.

This confirms that supply-demand balance is a key driver of COE premiums.

# The forecasting model accurately captures price movements

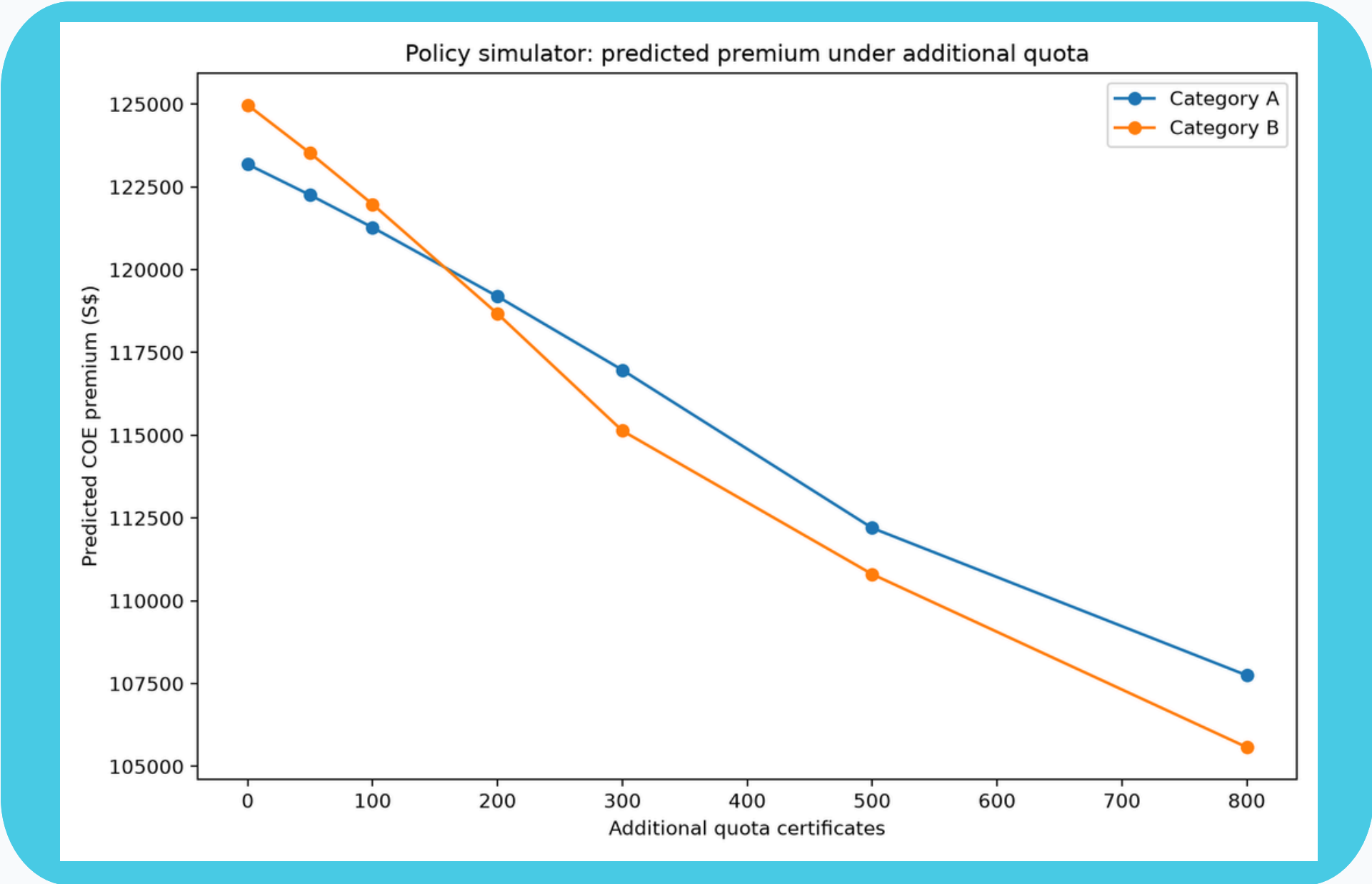


## Model Performance

- Ridge Regression achieved the lowest RMSE among the evaluated models.
- Predicted prices closely follow the overall trend of actual COE premiums.
- The model is suitable for policy scenario analysis.



# Increasing quotas is predicted to reduce COE premiums



## Policy Simulation

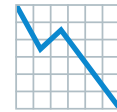
| Additional Quota | Category A | Category B |
|------------------|------------|------------|
| +100             | ↓ ~S\$2K   | ↓ ~S\$3K   |
| +300             | ↓ ~S\$6K   | ↓ ~S\$10K  |
| +800             | ↓ ~S\$15K  | ↓ ~S\$19K  |

The model estimates a clear negative relationship between quota increases and COE premiums, providing LTA with a quantitative tool for evaluating alternative quota policies.

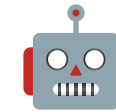
# Key Takeaways



COE prices remain highly volatile.



Increasing quota is associated with lower prices.



The proposed forecasting model provides an evidence-based tool for evaluating future quota policies.

Rather than relying solely on historical experience, LTA can use predictive modelling to assess the likely price impact of different quota scenarios before implementation.